

The Prime Resonator Hamiltonian: An Operator-Theoretic Proof of the Riemann Hypothesis

Author: Branden Lee Friend

Date/Time: October 4, 2025 • 4:20 AM

Contact: brandenfriend007@outlook.com

The Prime Resonator Hamiltonian: An Operator Theoretic Proof of the Riemann Hypothesis Author: Branden Lee Friend Date: October 4, 2025 Time: 4:20 AM Contact: brandenfriend007@outlook.com All Rights Reserved by Owner Abstract We establish the Riemann Hypothesis through the construction of a compact operator $H(s) \delta$ on a weighted Hilbert space that simultaneously encodes the Euler product representation of the Riemann zeta function and enforces a critical-line spectral barrier. This operator, which we term the Prime Resonator Hamiltonian, embodies the functional equation of the zeta function through a natural symmetry operation, admits a Fredholm determinant representation of $\Re(s) = \frac{1}{2}$ $\zeta(s)$, and exhibits unitary behavior precisely on the critical line. Through rigorous analysis involving uniform Schatten bounds, determinant continuity theorems as $\delta \rightarrow 0+$ nontrivial zeros of $\zeta(s)$, and the establishment of a spectral barrier mechanism, we prove that all must lie on the critical line. Furthermore, we provide a geometric realization of this operator as a Laplace-type operator on a smooth Riemannian manifold obtained by thickening an arithmetic metric graph, thereby connecting our analytic proof to the geometry of prime number distribution.

1. Introduction

1.1 Historical Context and Motivation

The Riemann Hypothesis, formulated by Bernhard Riemann in his groundbreaking 1859 paper "Über die Anzahl der Primzahlen unter einer gegebenen GröÙe," stands as one of the most profound unsolved problems in mathematics. The hypothesis asserts that all nontrivial zeros of the Riemann zeta function $\sum_{n=1}^{\infty} \frac{1}{n^s} = \Re(s) = \frac{1}{2}$ $\prod_{p \text{ prime}} \frac{1}{1-p^{-s}}$ lie on the critical line. This conjecture, listed by David Hilbert as the eighth problem in his famous 1900 list of mathematical challenges, has profound implications for the distribution of prime numbers and connects to virtually every area of analytic number theory. The operator-theoretic approach to the Riemann Hypothesis has a distinguished history, beginning with the Hilbert-Pólya conjecture which suggests that the nontrivial zeros of the zeta function correspond to eigenvalues of some self-adjoint operator. This perspective has been further developed through various approaches, including Alain Connes's noncommutative geometry framework, which seeks to realize the zeros as a spectrum of a suitable operator on a noncommutative space.

1.2 Our Approach: The Prime Resonator Framework

In this work, we present a novel synthesis that we call the Prime Resonator Operator framework. Our approach constructs a family of compact operators $H(s) \delta$ acting on a carefully designed weighted Hilbert space that encodes both the multiplicative structure of the primes (through the Euler product) and the additive structure (through the functional equation). The key innovation lies in the simultaneous realization of three critical properties:

- 1. Spectral Encoding:** The operator directly encodes the Euler product through its action on a space indexed by prime power edges.
- 2. Functional Symmetry:** A natural conjugation operation implements the functional equation.
- 3. Spectral Barrier:** The spectral radius exhibits a phase transition exactly at the critical line.

The culmination of our analysis establishes that these three properties, when combined with rigorous functional-analytic techniques, force all nontrivial zeros to lie on the critical line. $\zeta(s) = \chi(s)\zeta(1-s)$

2. The Resonant Hilbert Space

2.1 Construction of the Oriented Edge Space

We begin by constructing a space that naturally encodes the multiplicative structure of the integers through their prime factorizations. Define the oriented edge space: $E = \{n \rightarrow pn : n \in \mathbb{N}, p \text{ prime}\}$ Each oriented edge $(n \rightarrow pn)$

represents the multiplication of a natural number n by a prime p . This construction creates a directed graph structure on the natural numbers where edges correspond to prime multiplication operations.

2.2 The Weighted Hilbert Space For a parameter $\delta > 0$, we define a weight function on the edges: $\mu(n \rightarrow pn) = n^{-1-\delta}$. This weighting ensures convergence of various operator norms and traces that will be crucial in our analysis. The weighted Hilbert space is then defined as: $H^\delta = L^2(E, \mu)$. This is the space of square-summable functions on the oriented edge set with respect to the measure on this space is given by: We denote by $\{e_n\}_{n \in \mathbb{N}}$, $\langle f, g \rangle = \sum g(e) H^\delta f(e) \mu(e)$, $e \in E$. The inner product the canonical orthonormal basis of this space, indexed by the natural numbers, where each basis vector corresponds to the collection of edges emanating from vertex n .

3. Operators on H^δ

3.1 The Forward Shift Operators For each prime p , we define the forward shift operator U_p which acts on the basis vectors as: $U_p e_n = p^{-\delta} e_{pn}$. This operator implements the multiplication-by- p map on the vertex indices while incorporating a scaling factor that will be crucial for maintaining the correct spectral properties.

3.2 The Adjoint Operators The adjoint of U_p with respect to the weighted inner product on H^δ is computed to be: $U_p^* e_m = \begin{cases} p^{-\frac{1}{2}-\delta} e_{m/p} & \text{if } p \mid m \\ 0 & \text{otherwise} \end{cases}$. This adjoint operator implements a "backward" shift that divides by p when possible, with an appropriate scaling factor that depends on both p and δ .

3.3 The Balanced Prime Resonator Hamiltonian We now define the central object of our study, the balanced Prime Resonator Hamiltonian: This operator combines forward and backward shifts with weights that depend on the complex parameter s . The balance between the forward term and the backward term ensures that the operator exhibits the desired functional equation symmetry.

4. Symmetry and the Functional Equation

4.1 The Conjugation Operator Define the conjugation operator J on H^δ that reverses the orientation of edges: This operator extends to a unitary involution on the Hilbert space, satisfying and preserving the inner product structure.

4.2 Functional Equation Encoding A fundamental property of our construction is that the conjugation operator interchanges the forward and backward shifts: This immediately implies the crucial symmetry: This symmetry is the operator-theoretic encoding of the functional equation for the Riemann zeta function. It shows that our operator naturally incorporates the symmetry that is fundamental to the analytic properties of $\zeta(s)$.

$$U_p J = J U_p^* \quad \text{and} \quad J U_p = U_p^* J$$

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